THE DERIVATIVES

DIFFERENTIAL CALCULUS

prepared by:

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Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

60%

EXAM

40%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

TOPIC OUTLINE

The Slope of a Line

The Average Slope of a Line

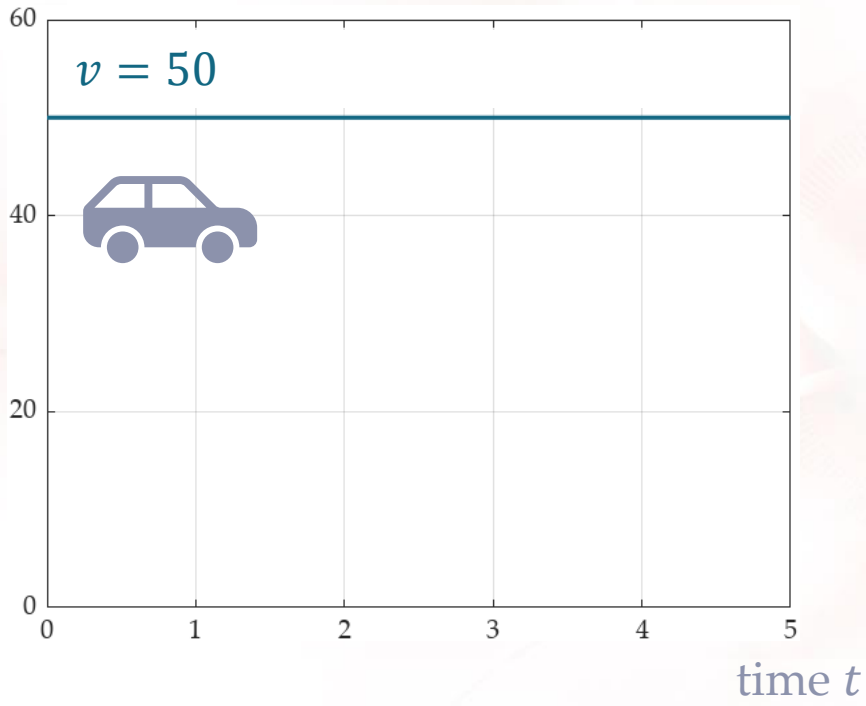
- Secant Line
- Tangent Line

The Derivatives

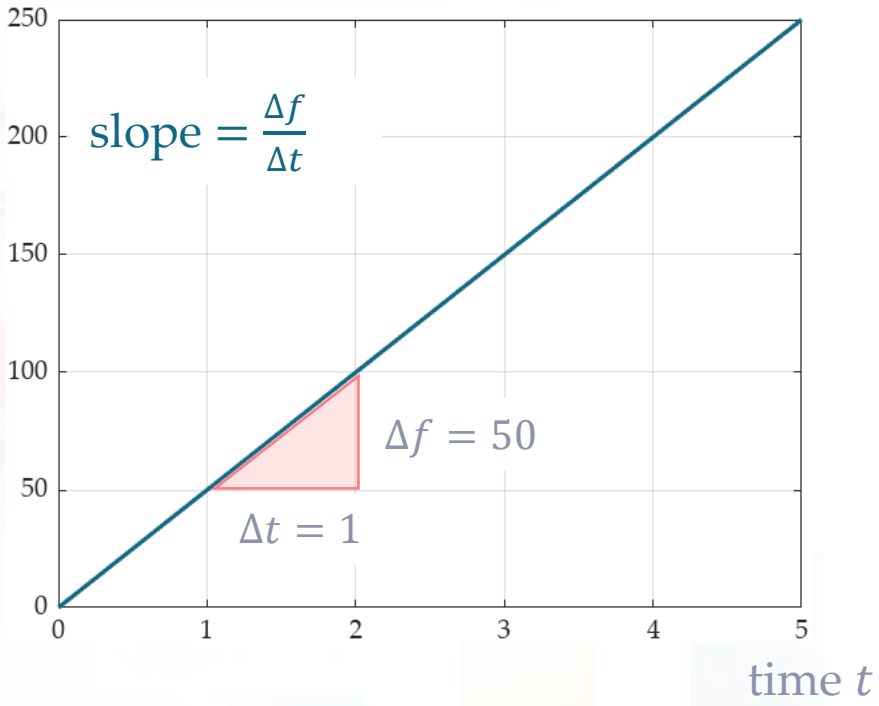
THE SLOPE OF A LINE

CONSTANT VELOCITY

velocity $v(t)$



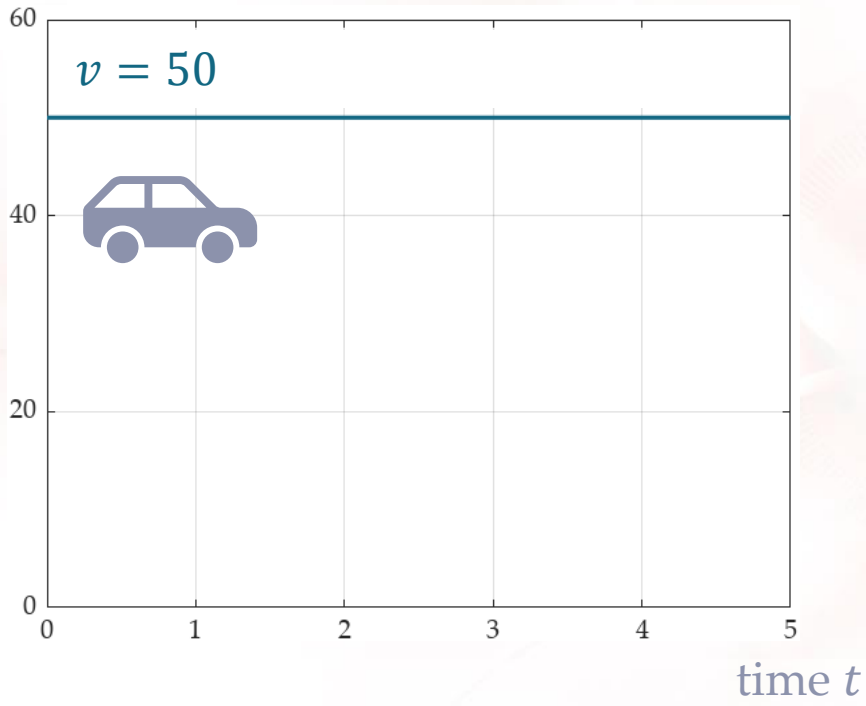
distance $f(t)$



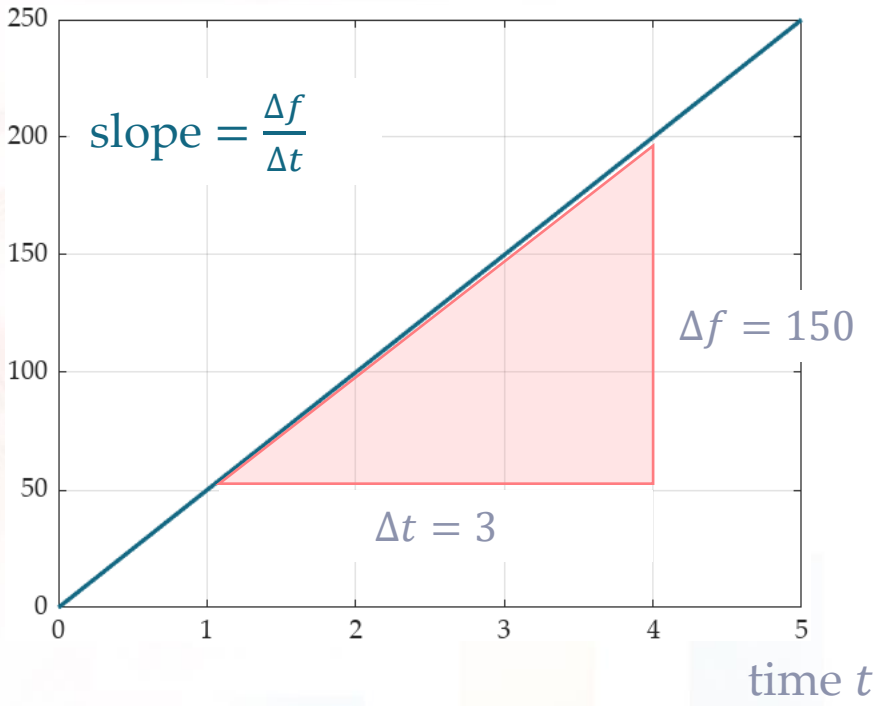
Linearly increasing distance $f(t) = 50t$

CONSTANT VELOCITY

velocity $v(t)$



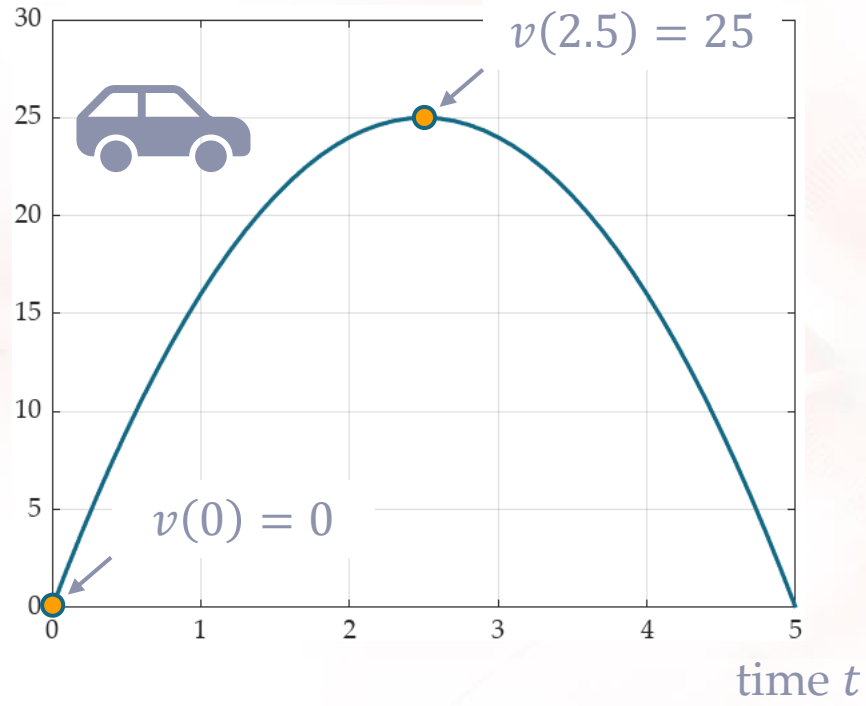
distance $f(t)$



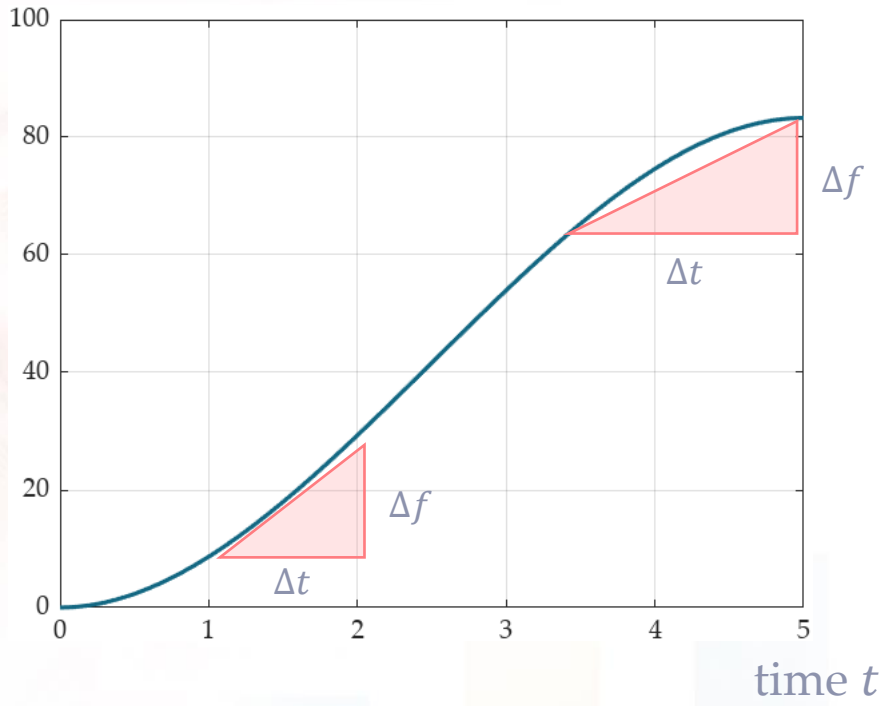
Linearly increasing distance $f(t) = 50t$

INSTANTENEOUS VELOCITY

velocity $v(t)$



distance $f(t)$

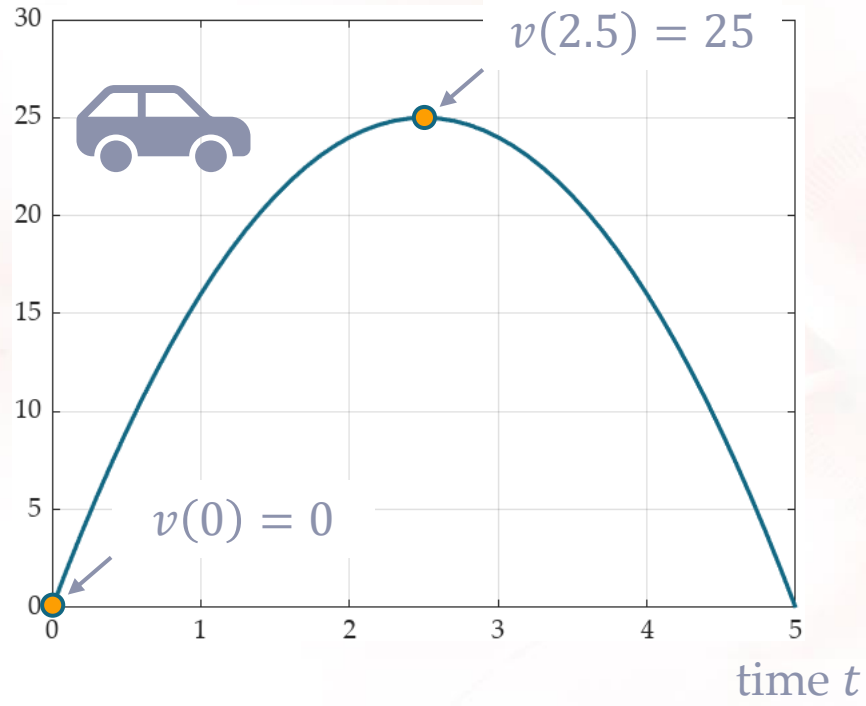


Real-life problems are nonlinear

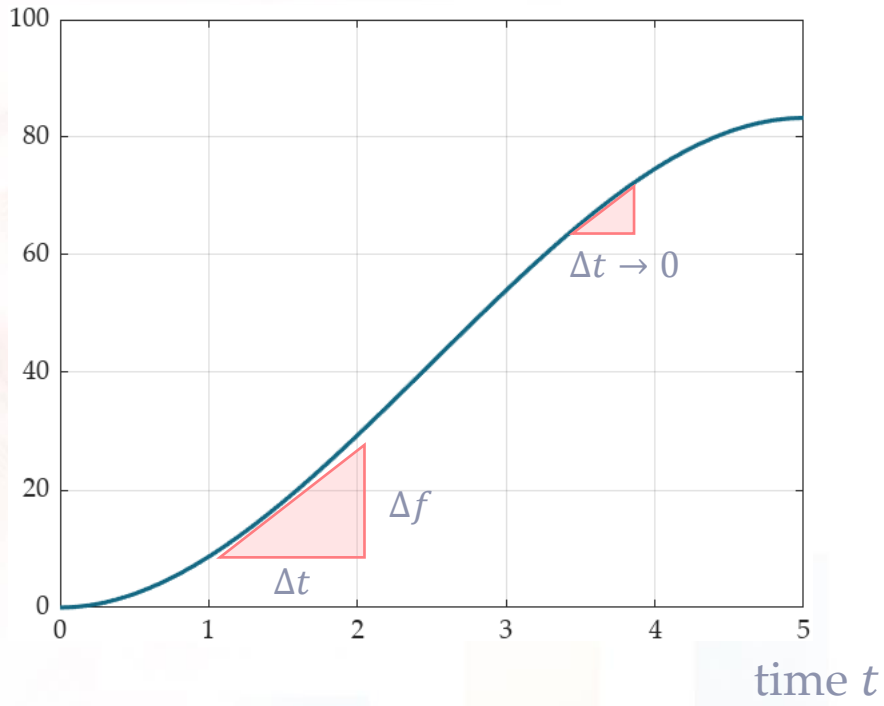
$\frac{\Delta f}{\Delta t}$ are not equal

INSTANTENEOUS VELOCITY

velocity $v(t)$



distance $f(t)$

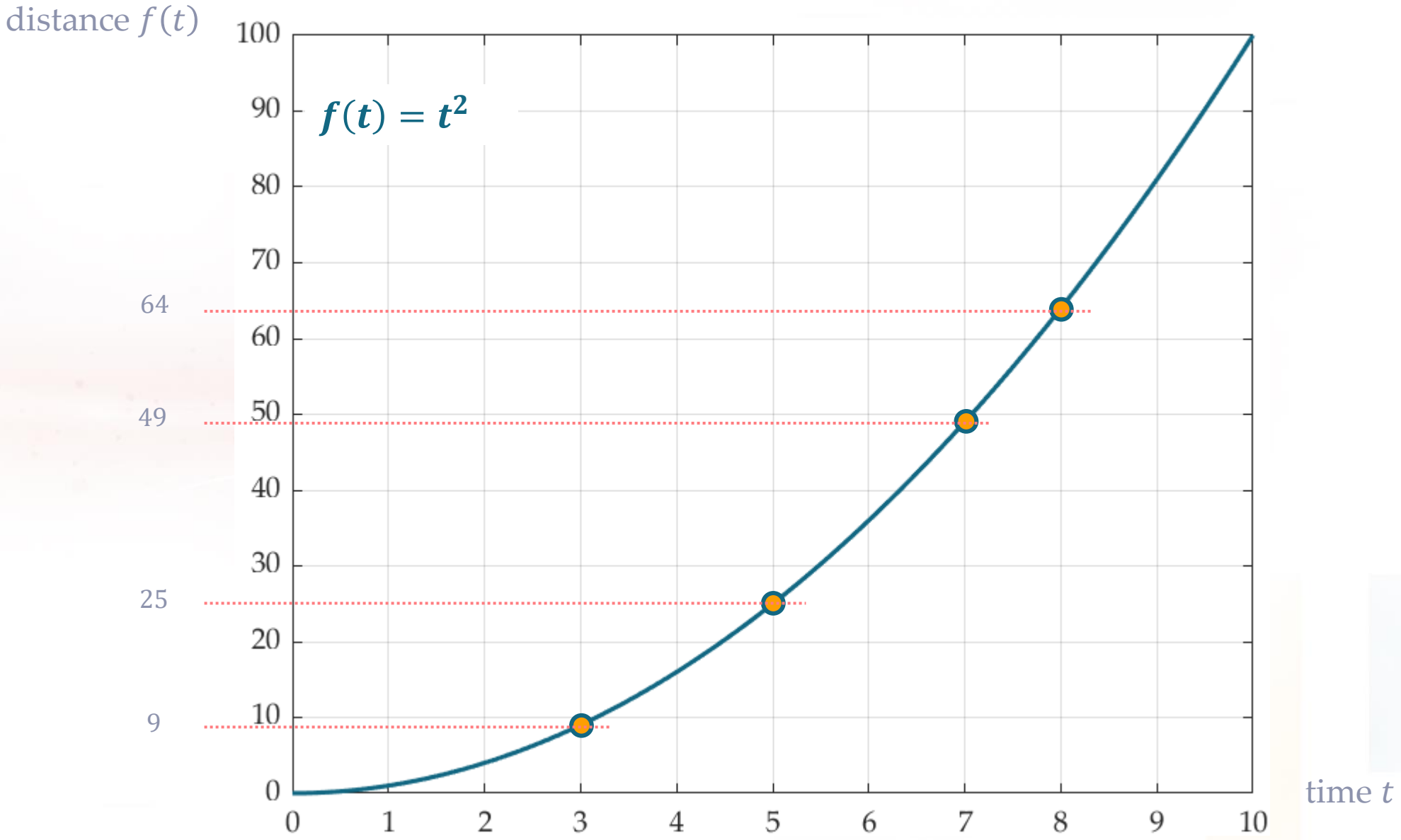


Real-life problems are nonlinear

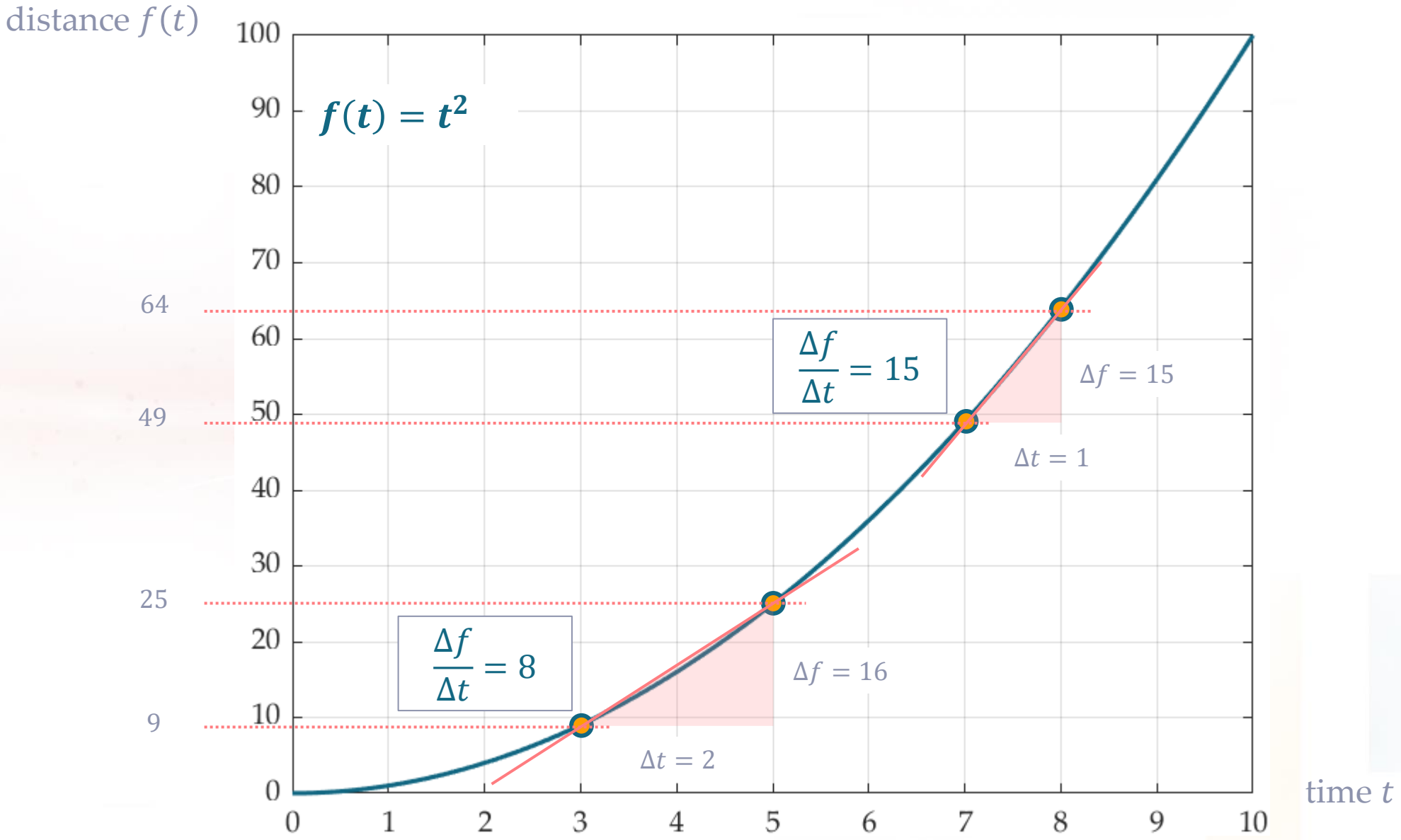
$\frac{\Delta f}{\Delta t}$ are not equal

THE AVERAGE SLOPE OF A LINE

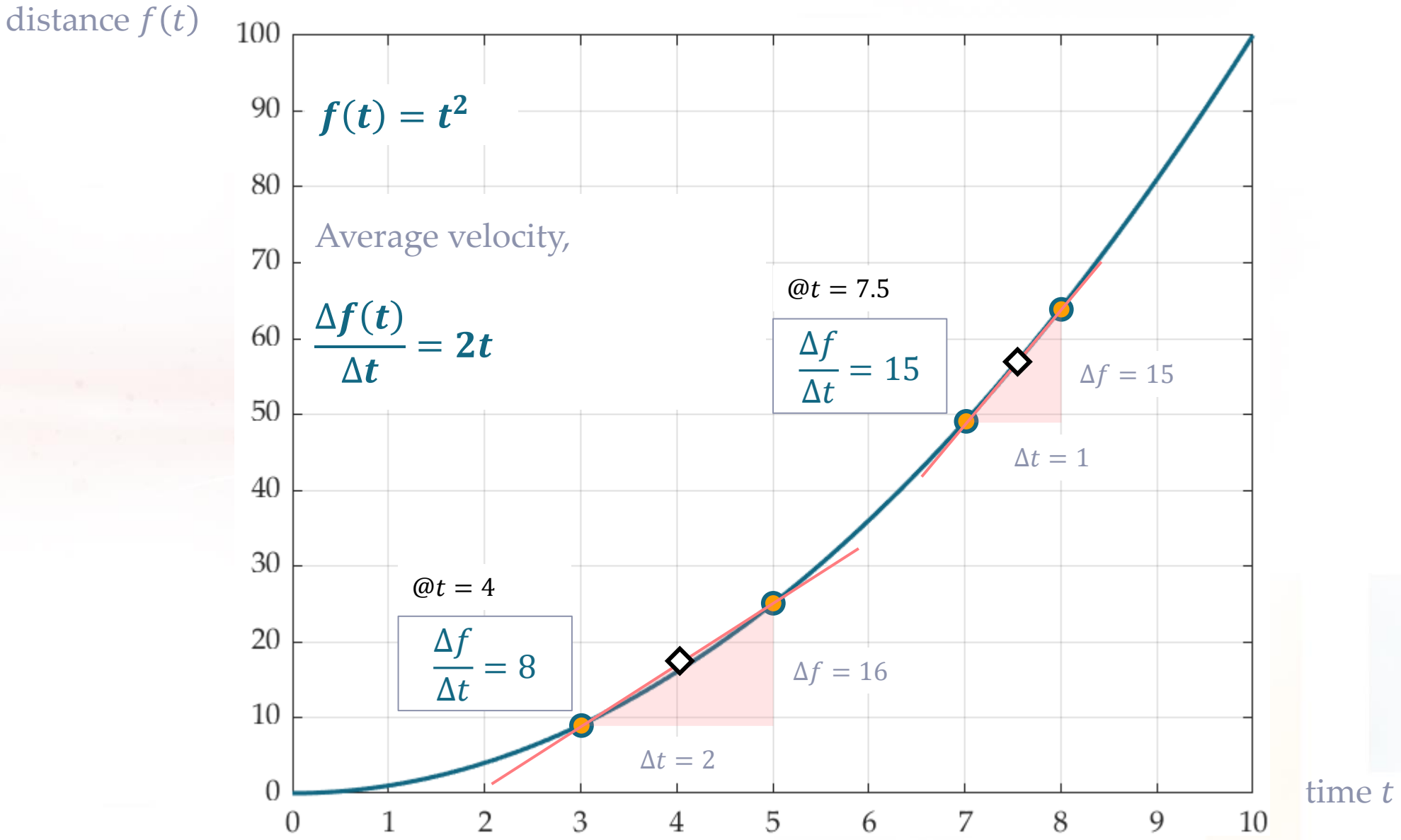
DISTANCE TRAVELED BY TIME t



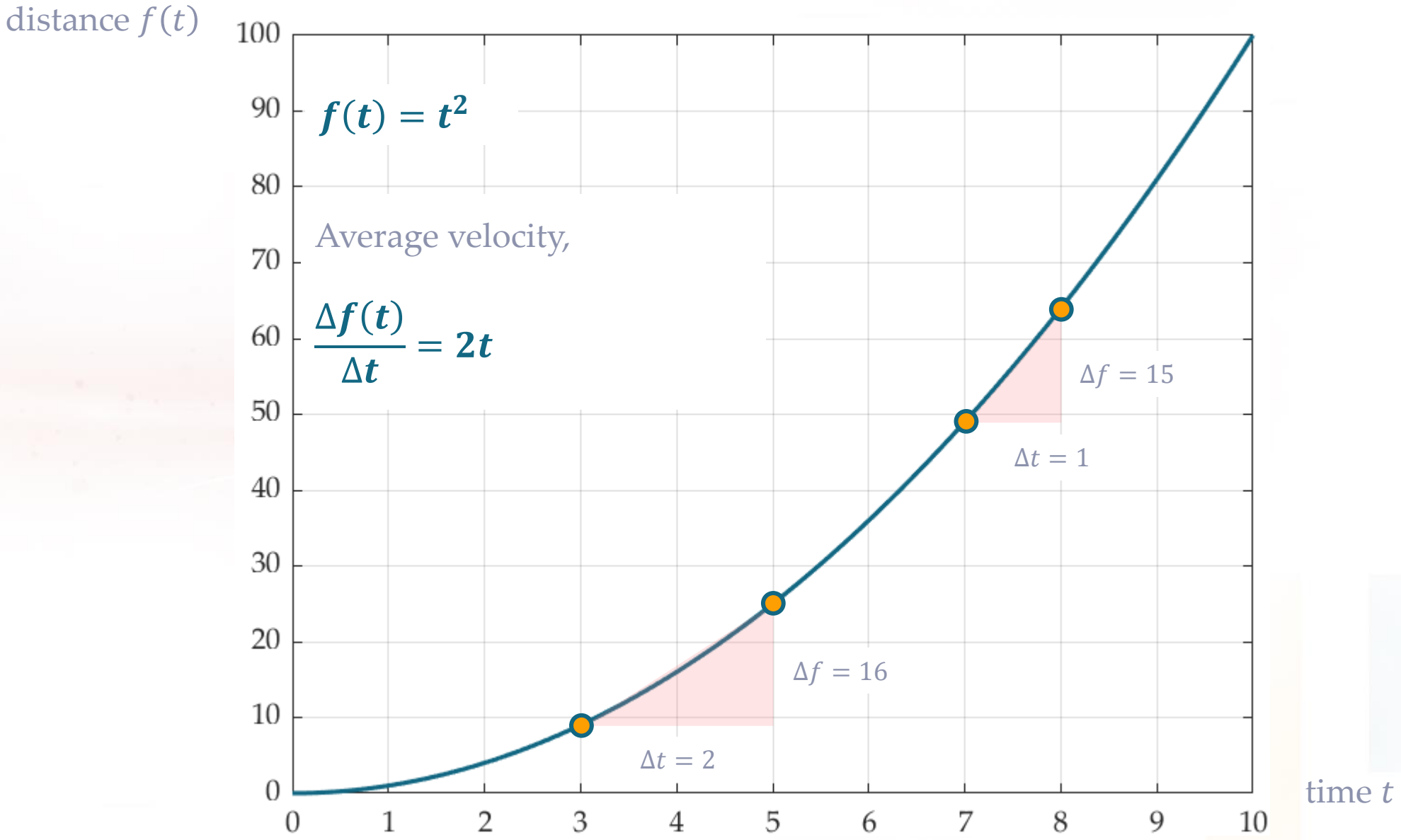
THE SECANT LINE



AVERAGE VELOCITY



THE VELOCITY AT AN INSTANT



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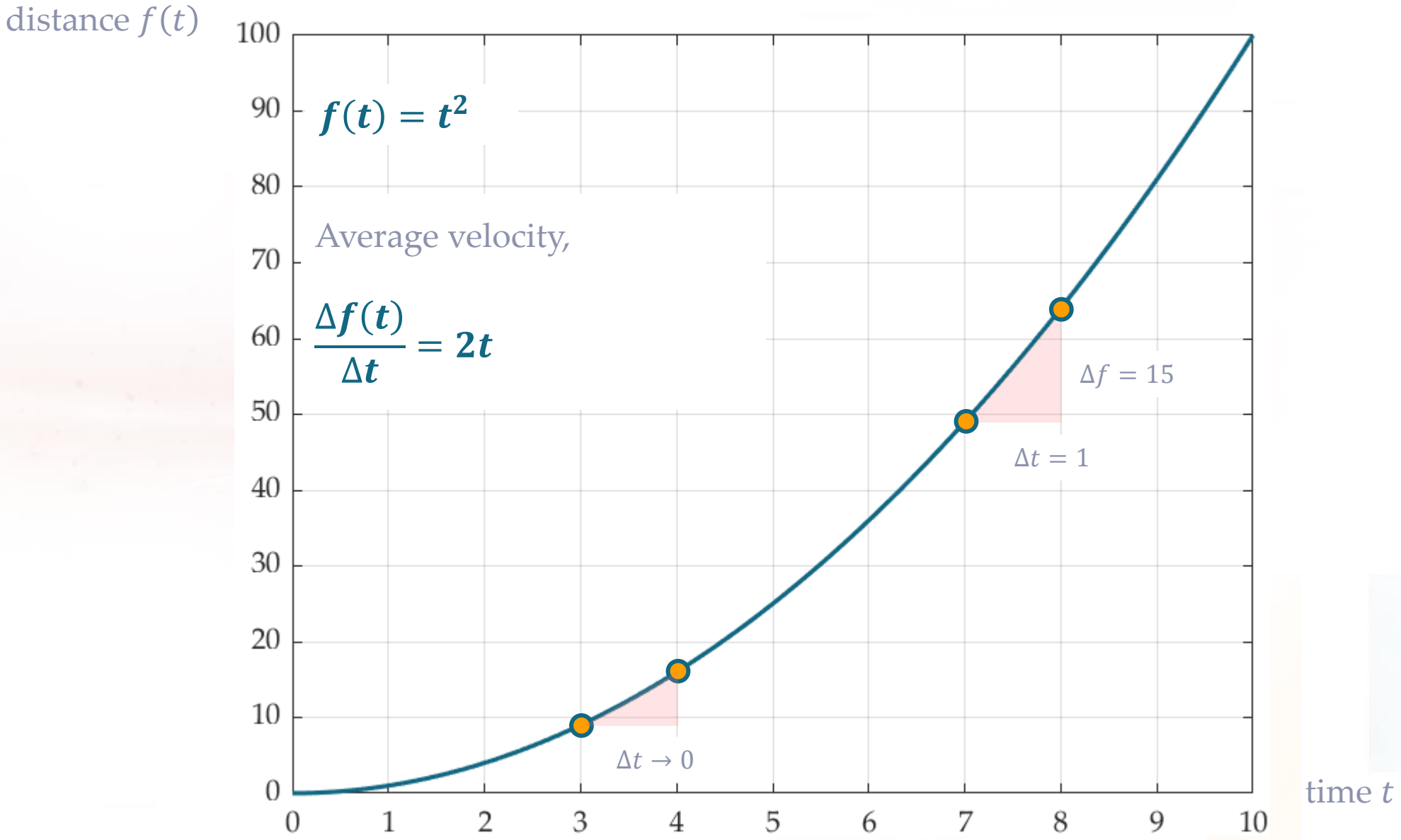
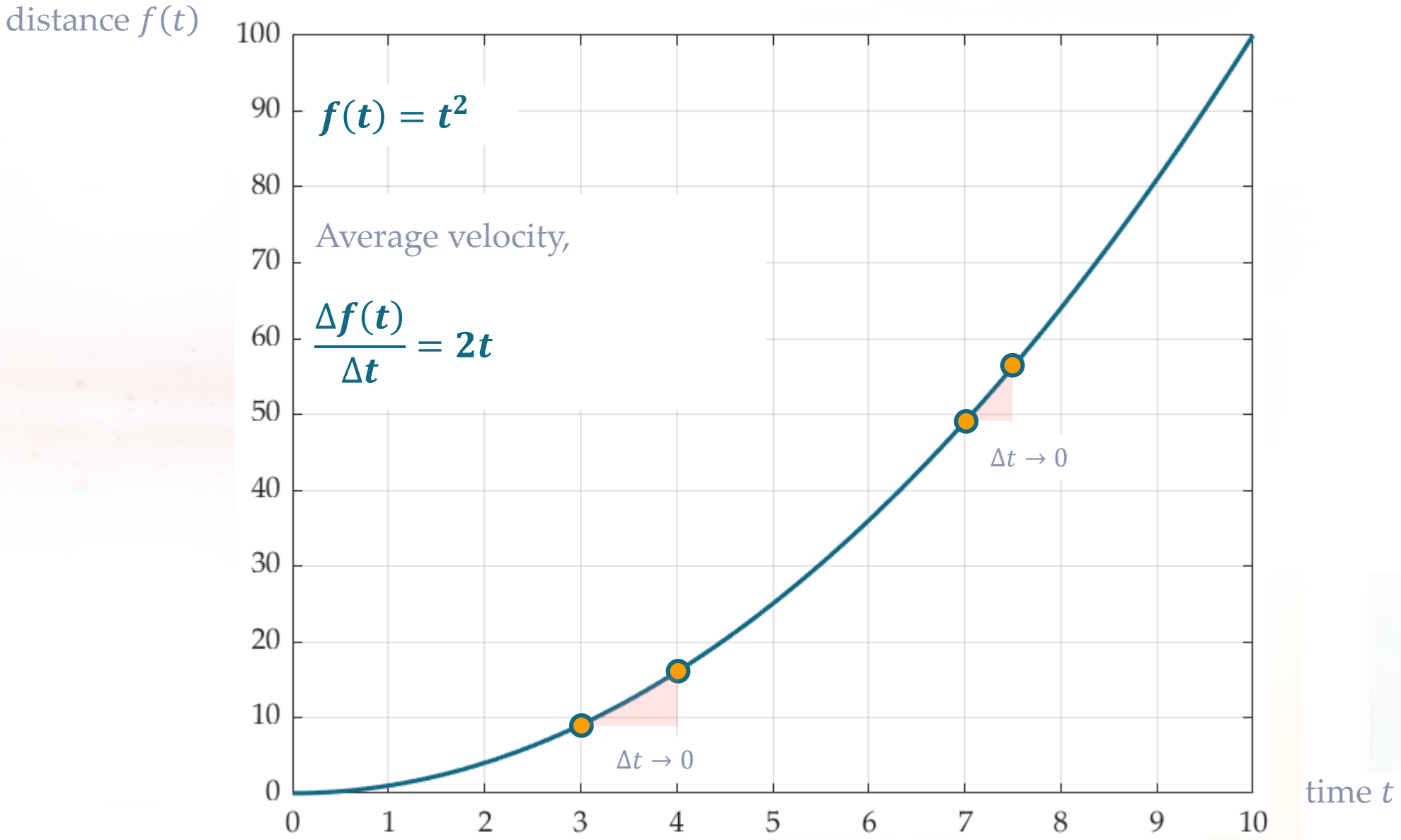
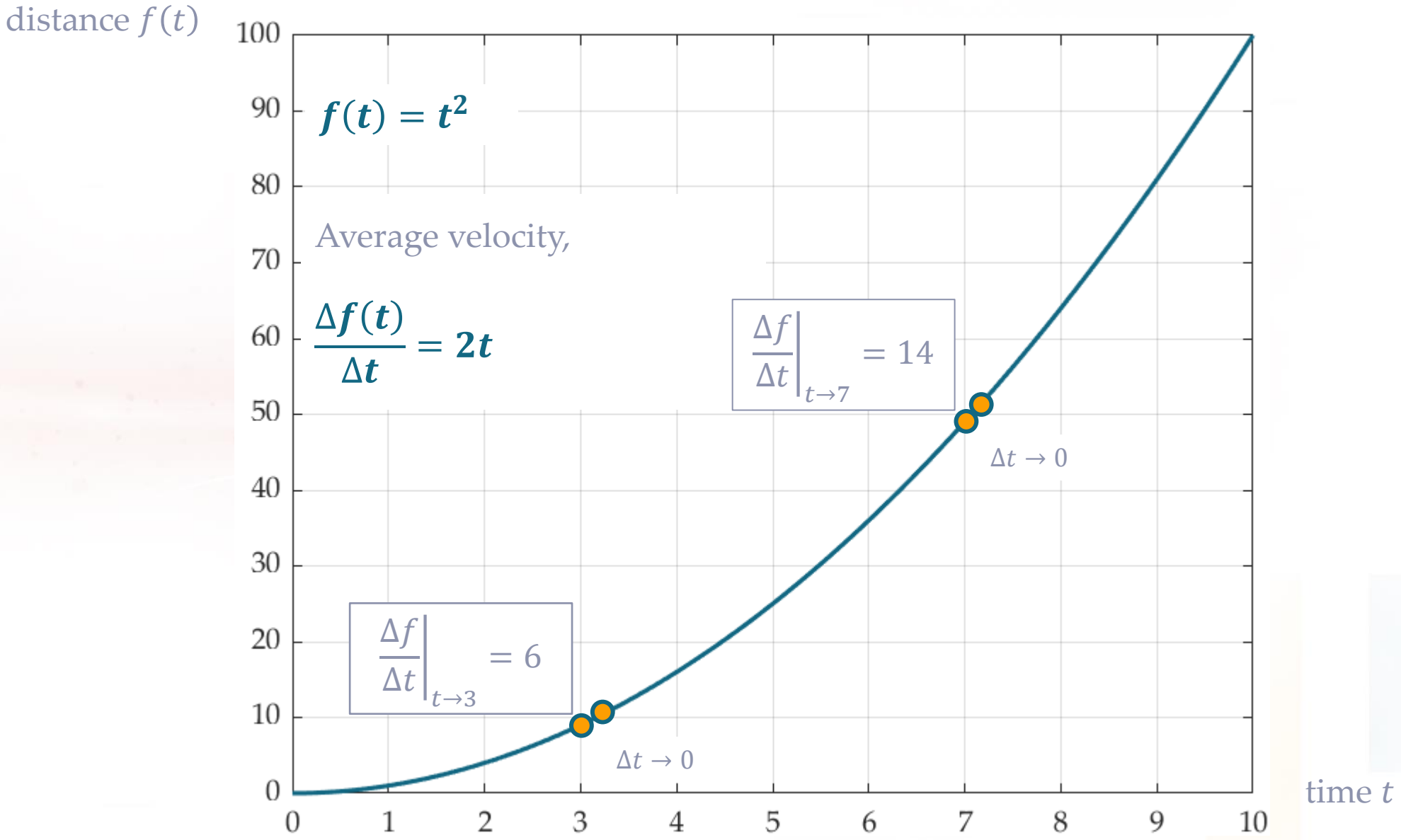
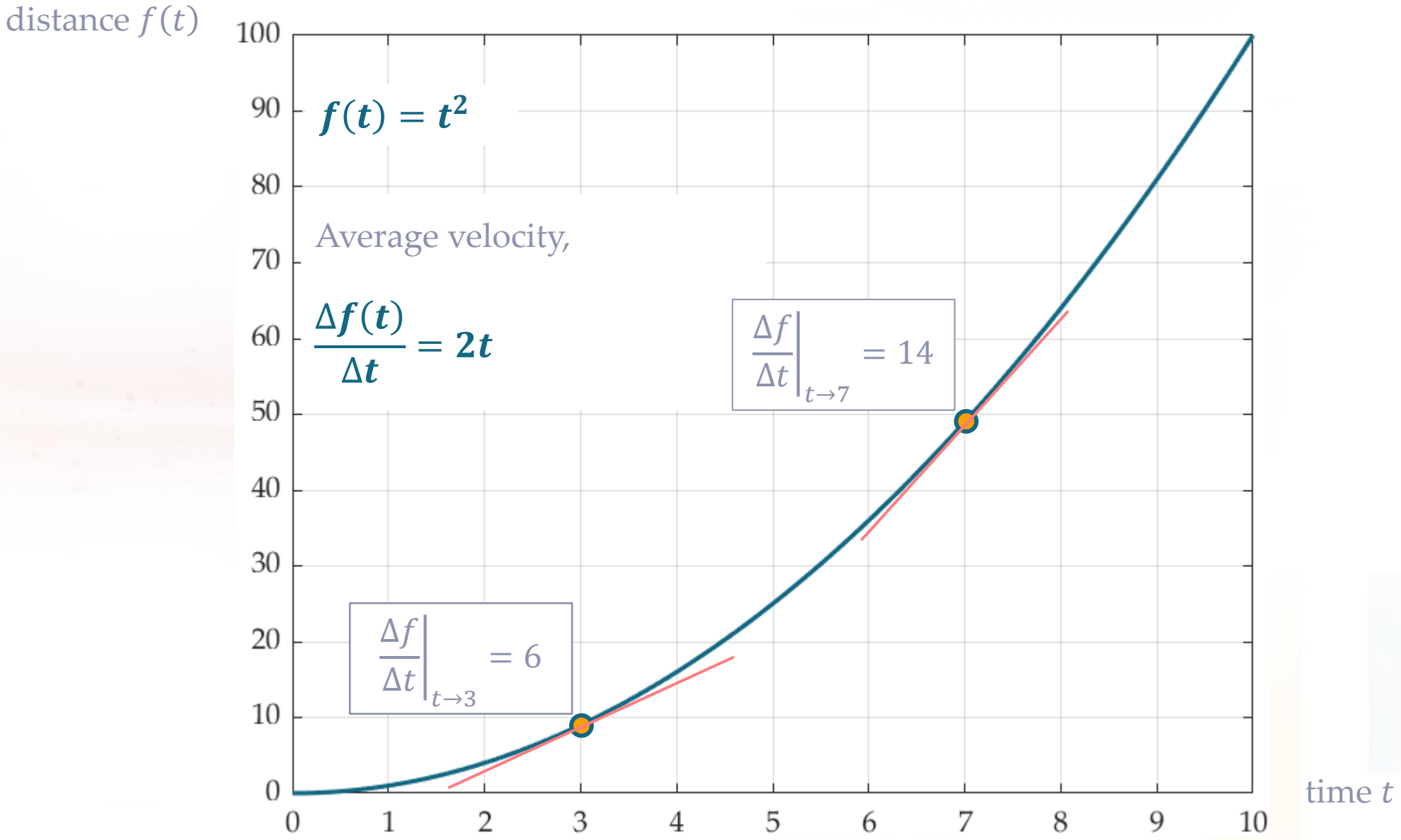


THE VELOCITY AT AN INSTANT



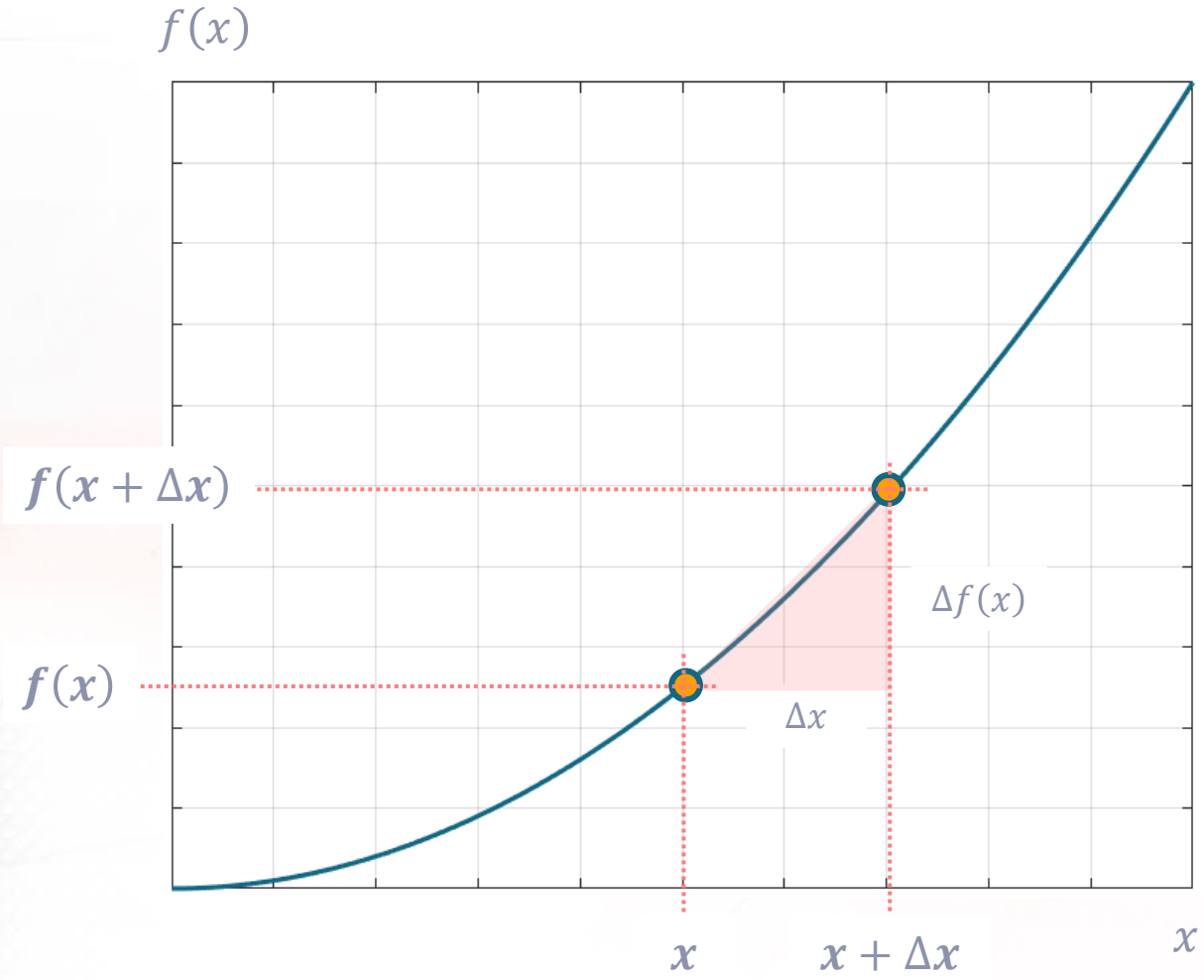
THE VELOCITY AT AN INSTANT




distance $f(t)$
100
90
80
70
60
50
40
30
20
10
0
$f(t) = t^2$
Average velocity,
$\frac{\Delta f(t)}{\Delta t} = 2t$
$\left.\frac{\Delta f}{\Delta t}\right|_{t\to 7} = 14$
$\left.\frac{\Delta f}{\Delta t}\right|_{t\to 3} = 6$
0
1
2
3
4
5
6
7
8
9
10
time t

THE DERIVATIVES

DERIVATIVE OF A FUNCTION



Difference Quotient

$$\frac{\Delta f(x)}{\Delta x} = \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

The Derivative of $f(x)$

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

Find the derivative of the function

$$f(x) = 2x$$

(use the difference quotient formula).

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$f'(x) = \lim_{\Delta x \to 0} \frac{2(x + \Delta x) - 2x}{\Delta x}$$

$$f'(x) = \lim_{\Delta x \to 0} \frac{2x + 2\Delta x - 2x}{\Delta x}$$

$$f'(x) = \lim_{\Delta x \to 0} \frac{2\Delta x}{\Delta x}$$

$$f'(x) = \lim_{\Delta x \to 0} 2$$

$$\boxed{f'(x) = 2}$$

ans

Find the derivative of the function

$$f(x) = 2x^2$$

(use the difference quotient formula).

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$f'(x) = \lim_{\Delta x \to 0} \frac{2(x + \Delta x)^2 - 2x^2}{\Delta x}$$

$$f'(x) = \lim_{\Delta x \to 0} \frac{2\left(x^2 + 2x\Delta x + (\Delta x)^2\right) - 2x^2}{\Delta x}$$

$$f'(x) = \lim_{\Delta x \to 0} \frac{2x^2 + 4x\Delta x + 2(\Delta x)^2 - 2x^2}{\Delta x}$$

$$f'(x) = \lim_{\Delta x \to 0} \frac{4x\Delta x + 2(\Delta x)^2}{\Delta x}$$

$$f'(x) = \lim_{\Delta x \to 0} \frac{2\Delta x(2x + \Delta x)}{\Delta x}$$

$$f'(x) = \lim_{\Delta x \to 0} 2(2x + \Delta x)$$

$$\boxed{f'(x) = 4x}$$

ans

Find the derivative of the function

$$f(x) = x^3 - x$$

(use the difference quotient formula).

$$f'(x) = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$f'(x) = \lim_{\Delta x \to 0} \frac{\left[(x + \Delta x)^3 - (x + \Delta x)\right] - \left(x^3 - x\right)}{\Delta x}$$

$$f'(x) = \lim_{\Delta x \to 0} \frac{(x + \Delta x)\left(x^2 + 2x\Delta x + (\Delta x)^3\right) - (x + \Delta x) - \left(x^3 - x\right)}{\Delta x}$$

$$f'(x) = \lim_{\Delta x \to 0} \frac{x^3 + 2x^2\Delta x + x(\Delta x)^3 + x^2\Delta x + 2x(\Delta x)^2 + (\Delta x)^4 - x - \Delta x - x^3 + x}{\Delta x}$$

$$f'(x) = \lim_{\Delta x \to 0} \frac{x^3 + 2x^2\Delta x + x(\Delta x)^3 + x^2\Delta x + 2x(\Delta x)^2 + (\Delta x)^4 - x - \Delta x - x^3 + x}{\Delta x}$$

$$f'(x) = \lim_{\Delta x \to 0} \frac{2x^2\Delta x + x(\Delta x)^3 + x^2\Delta x + 2x(\Delta x)^2 + (\Delta x)^4 - \Delta x}{\Delta x}$$

$$f'(x) = \lim_{\Delta x \to 0} \frac{\Delta x\left[2x^2 + x(\Delta x)^2 + x^2 + 2x\Delta x + (\Delta x)^3 - 1\right]}{\Delta x}$$

$$f'(x) = \lim_{\Delta x \to 0} 2x^2 + x(\Delta x)^2 + x^2 + 2x\Delta x + (\Delta x)^3 - 1$$

$$f'(x) = 2x^2 + x^2 - 1$$

$$\boxed{f'(x) = 3x^2 - 1}$$

ans

SEATWORK